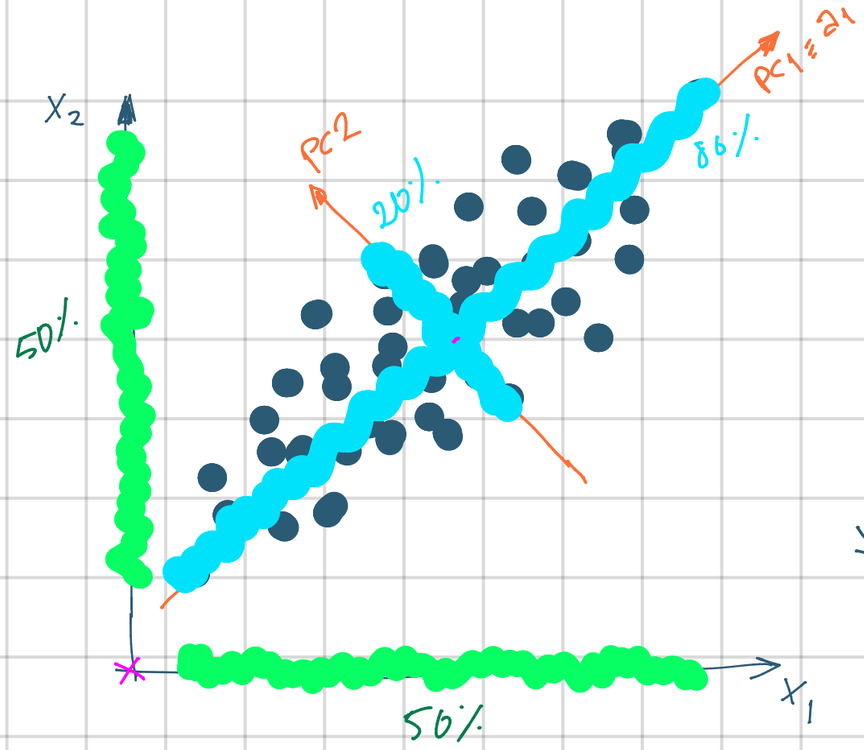


$$\begin{bmatrix} x_1 & x_2 \\ \vdots & \vdots \end{bmatrix}_{n\times 2} \xrightarrow{T(X)} \begin{bmatrix} y_1 & y_2 \\ \vdots & \vdots \end{bmatrix}_{n\times 2}$$

$y_1 \perp y_2$; $Var(y_1) \geq Var(y_2)$

$\rightarrow y_i = X_c a_i$ → autovector ; X_c → matriz centralizada

$y_1 = PC1 \begin{cases} \begin{bmatrix} y_1 \end{bmatrix}_{n\times 1} = \begin{bmatrix} X \end{bmatrix}_{n\times 2} \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix}_{2\times 1} \end{cases}$ (PC1)

$$X_c = X - 1\mu$$

$$X_c = \begin{bmatrix} \end{bmatrix}_{n\times 2} - \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_{n\times 1} \cdot \begin{bmatrix} \bar{x}_1 & \bar{x}_2 \end{bmatrix}_{1\times 2}$$

$$X_c = \begin{bmatrix} x_{11} & x_{12} \\ \vdots & \vdots \end{bmatrix}_{n\times 2} - \begin{bmatrix} \bar{x}_1 & \bar{x}_2 \\ \vdots & \vdots \\ \bar{x}_1 & \bar{x}_2 \end{bmatrix}_{n\times 2}$$

$$X_c = \begin{bmatrix} x_{11}-\bar{x}_1 & x_{12}-\bar{x}_2 \\ x_{21}-\bar{x}_1 & x_{22}-\bar{x}_2 \\ \vdots & \vdots \end{bmatrix}_{n\times 2}$$

$Y = X_c A \ldots (I)$; X_c (centrado) $\rightarrow \bar{y} = 0$

S: covarianza

$$Var(Y) = \frac{1}{n-1}\sum_{i=1}^{n}(y_i-\bar{y})^2 = \frac{1}{n-1}\sum_{i=1}^{n} y_i^2 = \frac{1}{n-1} Y^T Y = \frac{1}{n-1}(X_c A)^T (X_c A) = \frac{1}{n-1}(A^T X_c^T)(X_c A)$$

$Var(Y) = A^T S A \Rightarrow f(a) = a^T S a$; $g(a) = a^T a = 1$

"Truco escala" → Fijar la norma → buscamos DIRECCIÓN no un vector con longitud.

OBJETIVO: $\max_a f(a) = \max_a (a^T S a)$; sujeto a $a^T a = 1$

Construyendo Lagrangiano: $\max f(a)$ s.a. $g(a) = 0$

$$\Rightarrow L(a,\lambda) = f(a) - \lambda g(a)$$

$$\nabla L = 0 \longrightarrow \begin{cases} \begin{bmatrix} \frac{\partial L}{\partial a} \\ \frac{\partial L}{\partial \lambda} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{cases}$$

$$\Rightarrow L(a,\lambda) = a^T S a - \lambda(a^T a - 1)$$

$$\begin{bmatrix} \frac{\partial L}{\partial a} \equiv 2Sa - 2\lambda a \\ \frac{\partial L}{\partial \lambda} = -(a^T a - 1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow \begin{matrix} 2Sa - 2\lambda a = 0 \\ a^T a = 1 \end{matrix}$$

$$\Rightarrow \boxed{Sa = \lambda a} \rightarrow \begin{cases} \lambda: \text{autovalor} \rightarrow \text{varianza explicada} \\ \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix}_{p\times 1} \text{; donde } p = \#\text{variables.} \\ a: \text{autovector} \rightarrow \text{dirección Nuevas componentes} \\ \begin{bmatrix} \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} & \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} \end{bmatrix} \\ \quad PC1 \quad PC2 \end{cases}$$

$Sa - \lambda a = 0$

$Sa - \lambda I a = 0$

$(S - \lambda I)a = 0$

$$\det\left[\begin{pmatrix} Var(x_1) & Cov(x_2 x_1) \\ Cov(x_1 x_2) & Var(x_2) \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}\right] = 0$$

$$\det\left[\begin{pmatrix} Var(x_1)-\lambda & Cov(x_1 x_2) \\ Cov(x_1, x_2) & Var(x_2)-\lambda \end{pmatrix}\right] = 0$$

$$[Var(x_1)-\lambda)][Var(x_2)-\lambda] - Cov^2(x_1,x_2) = 0$$

$\rightarrow \lambda = \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix}$

Tipo especial de optimización

→ max/min ; con restricción

por ejm: $f(x,y) = xy + 1$

↳ punto de silla $(0,0)$.

→ Si tiene restricción

$g(x,y) = x^2 + y^2 - 1 = 0$

→ multiplicadores Lagrange:

$L(x,y,\lambda) = f(x,y) - \lambda g(x,y)$

$L = xy + 1 - \lambda(x^2+y^2-1)$

$\nabla L = \bar{0}$

$$\begin{bmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial y} \\ \frac{\partial L}{\partial \lambda} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} y - 2\lambda x \\ x - 2\lambda y \\ -(x^2+y^2-1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x^2 + y^2 = 1$

$y = 2\lambda x \rightarrow y = 2\lambda(2\lambda y)$

$x = 2\lambda y \quad y = 4\lambda^2 y$

$y \neq 0 \quad 1 = 4\lambda^2$

$\rightarrow \lambda_1 = \frac{1}{2}$ y $\lambda_2 = -\frac{1}{2}$

$y = 2\left(\frac{1}{2}\right)x$ y $y = 2\left(-\frac{1}{2}\right)x$

$y = x$ y $y = -x$

